VF Asia Footwear Laboratory Testing Manual

Test Method for Measurement of the Strength of Heel Attachment	Test method #: VF- F09 Page 1 of 5
Date Issued: May,2019 Revision #: 1 Reason for revision: Rectify the unit of measure from kg to kgf for strength of material.	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure the strength of heel attachment in completed footwear

REFERENCE: SATRA TM 113

EQUIPMENTS:

1. Tensile testing machine meeting the requirements of ASTM D 638.
2. Devices for clamping the shoe at the forepart and at the heel near its tip so that the heel can pivot freely during the test.
   - A metal bar of diameter 5.5± 0.5 mm and minimum length of 100 mm
   - A stirrup, of minimum width 80mm, capable of supporting the bar at both ends
3. Caliper capable of measuring up to 0.01mm. Digital type is preferred.
4. Hand or electric drill with twist drill bits of diameter @ 1.5 ±0.1 mm and @ 7.0 ± 0.5 mm

MATERIALS: Assembled shoes with high heels.

SAMPLE PREPARATION:

1. Condition the sample for at least 48 hours prior to testing at the standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

REPETITIONS: Test at least 1 pair

TEST PROCEDURE:

Test should be conducted at controlled laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

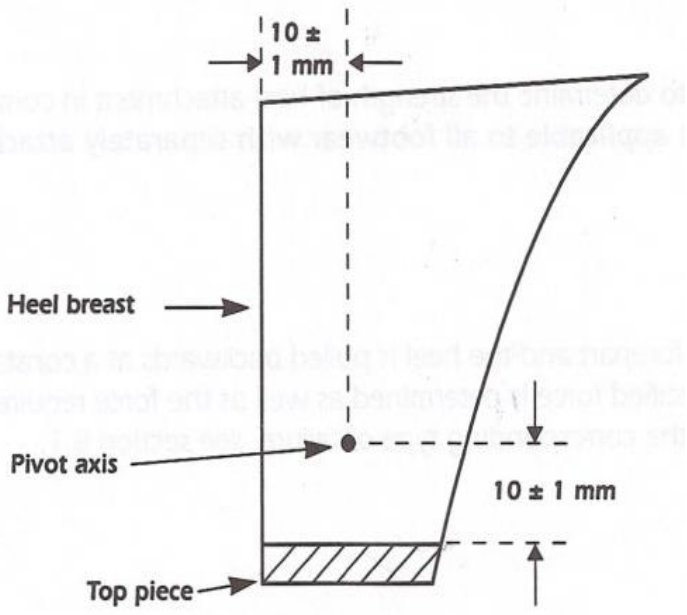
1. Equip the tensile testing machine with suitable jaws which are capable of clamping the test samples firmly in place such that no slippage occurs during testing.
2. Using the calipers, measure the heel height as the vertical distance between the top rear edge of the heel and the ground, with the top piece or heel base flat on the ground. Record this value as ***H.***
3. Determine and mark the location of the pivot point axis on the heel as shown in ***Figure 1.*** **-** If the heel is to be clamped using an inserted rod, (***refer to step# 4***), drill a hole aligned with the pivot axis of diameters just sufficient to allow free rotation of the rod.

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Figure 1

4. Clamp the shoe in the tensile testing machine using a rod inserted in the heel as shown in ***Figure 2a*** or an exterior heel clamp, as shown in ***Figure 2b***, for the following reasons:
   - It is possible to measure the distance between the heel and the forepart.
   - The plane of ground contact of the shoe and the longitudinal axis of the shoe back-part are aligned with the axis of the tensile machine, as shown in ***Figures 2a and 3***
   - The edge of the clamp grips the forepart a short distance forward of the front end of the shank
   - Zero force is applied to the shoe.





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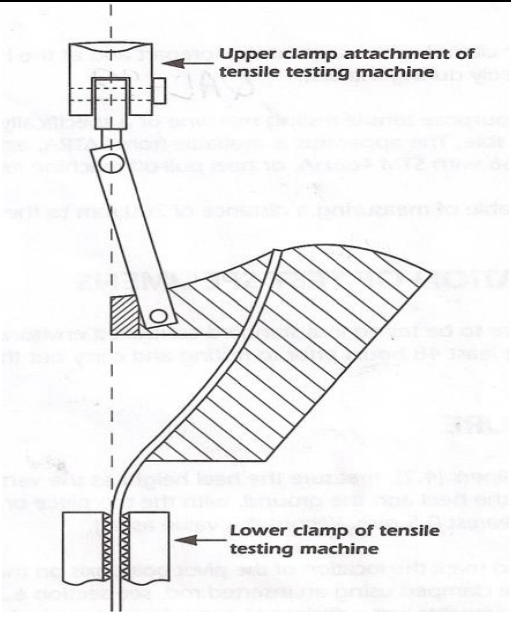
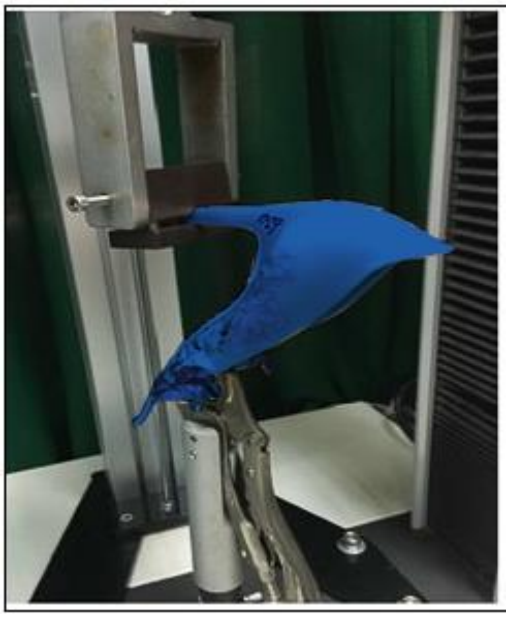


Figure 2a



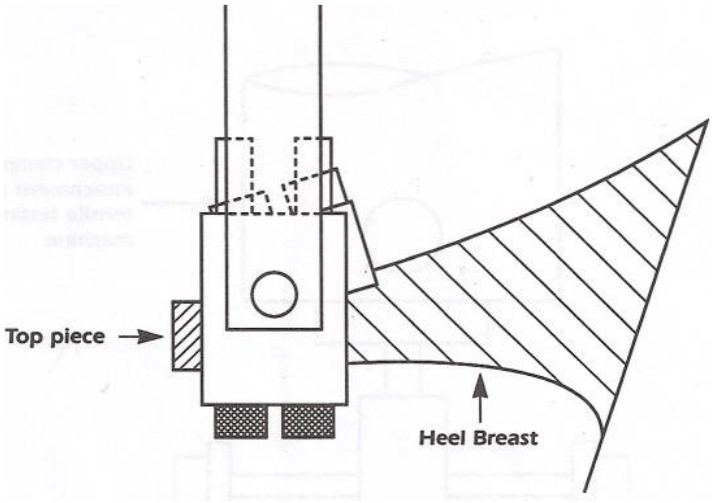


Figure 2b

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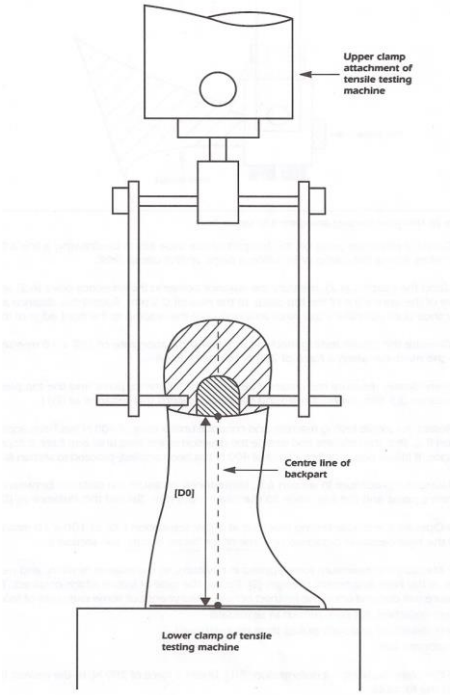


Figure 3

5. Define a reference point on the forepart of the shoe either by drawing a line few millimeters above the clamp or by using a point on the clamp itself.
6. Using the calipers, measure the distance between the reference point and the center of the front edge of the top piece, to the nearest 0.5 mm. Record this distance as ***DO.*** If the shoe does not have a top piece attached, take a reading to the front edge of the heel.
7. Operate the tensile testing machine at a jaw separation of 100mm/minute. Stop the machine when a force of 20 kgf has been applied.
8. Immediately measure the distance between the reference point and the top piece to the nearest 0.5 mm, without releasing the force. Record this distance as ***D1***
9. Restart the tensile machine and continue until the force of 40kgf has been applied ***(Refer to footnote 1)***. Stop the machine and reverse the direction of the jaws until zero force is applied to the shoe. If failure occurs before a force of 40kgf has been applied, proceed to step 11.
10. Repeat step 6. Record this distance as ***D2.***
11. Operate the tensile machine at a jaw separation of 100mm/minute until the heel becomes detached or some either failure occur. ***(Refer to footnote 2)***

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12. Measure the maximum force applied in kgf.

Notes on Back-Part Test:

1. ***The test method is intended to give a measure of the rigidity of the shoe back-part during walking, the deformation of the back-part caused by a reasonably large force applied to the heel in a backward direction, and the force required to detach the heel. @0kgf is two to three times the force applied to the heel during walking, however it is believed to be an appropriate level for predicting the back-part rigidity of the shoe in wear.***
2. ***In some cases, when the shank is weak, untampered or badly positioned, the back-part may bend considerably at the front of the heel without the heel becoming detached. In such cases there is usually an unsatisfactory large amount of permanent set after a force of 40kgf has been applied. It is usually necessary to apply a force of more than 1000 kgf in an attempt to produce heel detachment or some other type of failure.***

Calculation:

1. Back-part deformation, B1, under a force of 20kgf:
   B1 = (D1-D0)

2. Permanent set of the back-part, B2, after the application of a force of 40kgf:
   B2 = D2-D0

TEST REPORT:

1. Heel attachment strength. Maximum force in kgf
2. Type of failure
3. Optional:
   - B1: Back-part deformation
   - B2: Permanent set of the back-part